

Nana Kwesi Amponsah

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TECHNICAL SKILLS

Languages: Python, SQL, HTML/CSS, Matlab

Technologies: PyTorch, Docker, Git/GitHub, RAG, LangChain, APIs (OpenAI, Gemini, Anthropic, Grok), TensorFlow, NLP, Prompt Engineering, LLM Fine-tuning, HuggingFace Transformers, Keras, EDA, Data Visualization (Tableau), Data Warehousing, ETL, Machine Learning

PROJECTS

AI Research Agent | FastAPI, Docking, Grok API, JSON, PostgreSQL

[Project Link](#)

- Developed multi-agent AI research system with 4 specialized agents (Planner, Researcher, Writer, Editor) that autonomously conduct research and generate structured reports.
- Engineered full-stack solution using FastAPI, Docker, PostgreSQL, and Groq API with real-time progress tracking and agent orchestration
- Implemented ReAct pattern for iterative tool use, enabling the research agent to search across DuckDuckGo, arXiv, and Wikipedia with context-aware decision making

Multilingual Document Processor | RAG, LangChain, Transformers

[Project Link](#)

- Built a document Q&A and translation system supporting 12+ languages, leveraging transformer models: Flan-T5 for question answering and mBART-50 for multilingual translation.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline for context-aware question answering from PDFs.
- Orchestrated the workflow using LangChain: document loading, text chunking, embedding storage, retrieval, and LLM inference.

PROFESSIONAL EXPERIENCE

Beats by Dre (via Extern)- Remote

June 2025 – Sept 2025

Data Analytics Extern

- Conducted qualitative and quantitative research on Gen Z consumer behavior, uncovering insights on emerging audio-device preferences and trend shifts that informed early product-strategy discussions.
- Performed market and competitive analysis using Python, the Gemini API, and **NLTK-based sentiment-analysis pipelines** to extract and interpret consumer opinions at scale.
- Built a polished, insight-driven **final analytics presentation deck** using Gamma, incorporating **statistical visualizations** and narrative explanations generated with Claude AI's Artifact workspace.

Excelerate – Dubai, Remote

May 2025 - Aug 2025

Data Analyst Intern

- **Led and coordinated an international team of students**, guiding data cleaning workflows, feature engineering, and model development to ensure consistent, high-quality outputs across the project.
- Built and deployed predictive and recommendation models, including a Random Forest regressor achieving 98% accuracy in churn prediction and a lightweight recommendation system that improved student engagement pathways.

EDUCATION

Kwame Nkrumah University of Science & Technology

Jan 2021 - Sept 2024

Bachelor of Science, Mechanical Engineering

CERTIFICATIONS

IBM AI Engineering Professional Certificate | **IBM (via Coursera)**

Nov 2025

Agentic AI | **DeepLearning.AI**

Nov 2025